CORE PURE / MISCELLANEOUS PURE

Method of Differences

Mark scheme 12 questions · 90 marks

Questions

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	Total	**90**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$1+\frac{2}{r(r+3)}=\frac{r^2+3r+2}{r(r+3)}$ $=\frac{(r+1)(r+2)}{r(r+3)}$	**M1**	Forms a single fraction and factorises its numerator, or expands the product of the two RHS fractions. Condone one slip.
	$\ln\left(\frac{r+1}{r}\right)+\ln\left(\frac{r+2}{r+3}\right)$ $=\ln\left(\frac{r+1}{r}\right)-\ln\left(\frac{r+3}{r+2}\right)$	**A1 AG**	Correct fraction split into two logarithms and reaching the printed difference without error, via the reciprocal or $\ln a-\ln b$.
		[2]	
1(b)	$\left(\ln 2-\ln\frac{4}{3}\right)+\left(\ln\frac{3}{2}-\ln\frac{5}{4}\right)+\left(\ln\frac{4}{3}-\ln\frac{6}{5}\right)$ $+\cdots+\left(\ln\frac{n}{n-1}-\ln\frac{n+2}{n+1}\right)$ $+\left(\ln\frac{n+1}{n}-\ln\frac{n+3}{n+2}\right)$	**M1**	Uses part (a) with $r=1$, $r=2$, $r=3$, $r=n-1$ and $r=n$ at least, enough to show terms cancelling. Condone slips in individual terms.
	$\ln 2+\ln\frac{3}{2}-\ln\frac{n+2}{n+1}-\ln\frac{n+3}{n+2}$	**A1**	All four surviving terms correct and no others, oe such as $\ln 3$ for the first two.
	$\ln\left(2\times\frac{3}{2}\times\frac{n+1}{n+2}\times\frac{n+2}{n+3}\right)$	**M1**	Uses log laws to combine their surviving terms into a single logarithm. Condone sign slips.
	$=\ln\left(\frac{3(n+1)}{n+3}\right)$	**A1 AG**	Correct single logarithm simplified to the printed result without error.
		[4]	
Total for Question 1		**6**	

Question	Scheme	Marks	Notes
2(a)	$\frac{1}{r(r+2)} = \frac{A}{r} + \frac{B}{r+2} \Rightarrow 1 = A(r+2) + Br$ $\Rightarrow \frac{1}{r(r+2)} = \frac{1}{2r} - \frac{1}{2(r+2)}$	**M1 A1**	**M1** Uses the form $\frac{A}{r} + \frac{B}{r+2}$ and finds A and B by any method. Condone slips in their values. **A1** $\frac{1}{2r} - \frac{1}{2(r+2)}$, oe, such as $A = \frac{1}{2}$, $B = -\frac{1}{2}$ stated.
	$\frac{1}{2}\left[(1 - \frac{1}{3}) + (\frac{1}{2} - \frac{1}{4}) + (\frac{1}{3} - \frac{1}{5}) + \cdots\right.$ $\left. + (\frac{1}{n-1} - \frac{1}{n+1}) + (\frac{1}{n} - \frac{1}{n+2})\right]$	**dM1**	Writes their differences for at least $r = 1, 2, 3$ and $r = n-1, n$, showing terms cancelling. The factor $\frac{1}{2}$ need not be seen.
	$\frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right)$	**A1**	The four correct surviving terms, with the factor $\frac{1}{2}$.
	$\frac{1}{2}\left(\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right)$ $= \frac{3}{4} - \frac{1}{2(n+1)} - \frac{1}{2(n+2)}$	**A1 AG**	Printed result reached without error from correct partial fractions and surviving terms.
		[5]	
2(b)	$n \to \infty \Rightarrow \frac{1}{2(n+1)} \to 0,\ \frac{1}{2(n+2)} \to 0$	**M1**	Takes the limit as $n \to \infty$ in the result of (a), with both terms in n tending to 0.
	$\sum_{r=1}^{\infty} \frac{1}{r(r+2)} = \frac{3}{4}$	**A1**	
		[2]	
	Total for Question 2	**7**	

Question	Scheme	Marks	Notes
3	$\frac{1}{r(r+3)} = \frac{1}{3}\left(\frac{1}{r} - \frac{1}{r+3}\right)$	**M1**	Splits the general term into the form $\frac{A}{r} + \frac{B}{r+3}$ and finds A and B by any method. Condone slips in their values.
	$\frac{1}{3}\left[\left(1 - \frac{1}{4}\right) + \left(\frac{1}{2} - \frac{1}{5}\right) + \left(\frac{1}{3} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{n-2} - \frac{1}{n+1}\right) + \left(\frac{1}{n-1} - \frac{1}{n+2}\right) + \left(\frac{1}{n} - \frac{1}{n+3}\right)\right]$	**dM1**	Lists their differences for at least $r = 1, 2, 3$ and $r = n-2, n-1, n$, showing or implying the cancellation. Condone a missing $\frac{1}{3}$.
	$\frac{1}{3}\left(1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3}\right)$	**A1**	Correct surviving terms with the factor $\frac{1}{3}$, oe.
	$\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3}$ $= \frac{(n+2)(n+3)+(n+1)(n+3)+(n+1)(n+2)}{(n+1)(n+2)(n+3)}$	**M1**	Combines their fractions in n over any common denominator, with each numerator multiplied by the missing factors. Condone expansion slips.
	$\frac{11}{18} - \frac{3n^2+12n+11}{3(n+1)(n+2)(n+3)}$ $a = 3,\ b = 12,\ c = 11$	**A1**	cso. Values stated, or seen in the completed expression.
	Total for Question 3	**5**	

Question	Scheme	Marks	Notes
4	$\frac{12}{(2r-1)(2r+1)(2r+3)}$ $\equiv \frac{A}{2r-1}+\frac{B}{2r+1}+\frac{C}{2r+3}$ $\Rightarrow A=\ldots,\ B=\ldots,\ C=\ldots$	**M1**	Writes the term as three fractions over $2r-1$, $2r+1$, $2r+3$ and finds a constant by substitution, cover-up or comparing coefficients.
	$\frac{3}{2(2r-1)}-\frac{3}{2r+1}+\frac{3}{2(2r+3)}$	**A1**	oe, such as $A=\frac{3}{2}$, $B=-3$, $C=\frac{3}{2}$ stated.
	$\frac{3}{2}\left[\left(1-\frac{2}{3}+\frac{1}{5}\right)+\left(\frac{1}{3}-\frac{2}{5}+\frac{1}{7}\right)\right.$ $+\left(\frac{1}{5}-\frac{2}{7}+\frac{1}{9}\right)+\cdots$ $+\left(\frac{1}{2n-3}-\frac{2}{2n-1}+\frac{1}{2n+1}\right)$ $\left.+\left(\frac{1}{2n-1}-\frac{2}{2n+1}+\frac{1}{2n+3}\right)\right]$	**dM1**	Lists their terms for at least three values of r at the start and two at the end, or two and two after splitting into two differences, showing cancellation.
	$\frac{3}{2}\left(1-\frac{1}{3}-\frac{1}{2n+1}+\frac{1}{2n+3}\right)$	**A1**	Correct surviving terms oe; need not be collected.
	$1-\frac{3}{2(2n+1)}+\frac{3}{2(2n+3)}$ $=\frac{2(2n+1)(2n+3)-3(2n+3)+3(2n+1)}{2(2n+1)(2n+3)}$	**M1**	Combines their surviving terms, including at least two with denominators in n, over a common denominator containing $(2n+1)(2n+3)$. Condone slips in the numerator.
	$=\frac{n(4n+8)}{(2n+1)(2n+3)}$, $a=4$, $b=8$	**A1**	cso. Needs $a=4$ and $b=8$, or the sum written as $\frac{n(4n+8)}{(2n+1)(2n+3)}$; $\frac{4n(n+2)}{(2n+1)(2n+3)}$ alone scores A0.
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5(a)	$1+\frac{2}{r(r+3)}=\frac{(r+1)(r+2)}{r(r+3)}$ $\Rightarrow \ln(r+1)-\ln r+\ln(r+2)-\ln(r+3)$	**M1**	Forms a single fraction, factorises its numerator and splits the logarithm into logarithms of linear factors, or of two ratios. Condone sign slips.
	$r=1$: $\ln 2-\ln 1+\ln 3-\ln 4$ $r=2$: $\ln 3-\ln 2+\ln 4-\ln 5$ ... $r=n-1$: $\ln n-\ln(n-1)+\ln(n+1)-\ln(n+2)$ $r=n$: $\ln(n+1)-\ln n+\ln(n+2)-\ln(n+3)$	**dM1**	Lists their terms for at least $r=1$, $r=2$, $r=n-1$ and $r=n$, showing cancellation.
	$\ln(n+1)-\ln 1+\ln 3-\ln(n+3)$	**A1**	Correct surviving terms only; $\ln 1$ may be omitted.
	$\ln 3+\ln(n+1)-\ln(n+3)=\ln\left(\frac{3(n+1)}{n+3}\right)$	**M1** **A1 AG**	**M1** Uses log laws to combine their surviving terms into a single logarithm. **A1 AG** Printed result reached without error from correct surviving terms; nothing for working back from the result.
		[5]	
5(b)	$28\sum_{r=25}^{74}\ln\left(1+\frac{2}{r(r+3)}\right)$	**B1**	
	$\ln\left(\frac{3(75)}{77}\right)-\ln\left(\frac{3(25)}{27}\right)$	**M1**	Subtracts the result of (a) at $n=24$ from that at $n=74$.
	$28\ln\left(\frac{81}{77}\right)$	**A1**	Or another form $a\ln\left(\frac{b}{c}\right)$ with integers a, b, c. Not for a decimal or an uncombined difference of logarithms.
		[3]	
	Total for Question 5	**8**	

Question	Scheme	Marks	Notes
6	$\frac{6}{r(r+1)(r+2)} = \frac{A}{r(r+1)} + \frac{B}{(r+1)(r+2)}$ $\Rightarrow 6 = A(r+2) + Br \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Writes the term as $\frac{A}{r(r+1)} + \frac{B}{(r+1)(r+2)}$ and finds both constants by any method. Condone a slip in one constant.
	$\frac{6}{r(r+1)(r+2)} = \frac{3}{r(r+1)} - \frac{3}{(r+1)(r+2)}$	**A1**	
	$\left(\frac{3}{1\times2} - \frac{3}{2\times3}\right) + \left(\frac{3}{2\times3} - \frac{3}{3\times4}\right)$ $+ \cdots + \left(\frac{3}{n(n+1)} - \frac{3}{(n+1)(n+2)}\right)$	**dM1**	Writes out their partial fractions for at least $r = 1$, $r = 2$ and $r = n$, showing terms cancelling.
	$\frac{3}{2} - \frac{3}{(n+1)(n+2)}$	**A1**	Correct surviving terms, oe.
	$\frac{3(n+1)(n+2)-6}{2(n+1)(n+2)} = \frac{n(\ldots)}{2(n+1)(n+2)}$	**M1**	Combines their $p - \frac{q}{(n+1)(n+2)}$ over a common denominator and simplifies the numerator. Condone slips in expanding.
	$\frac{3n^2+9n}{2(n+1)(n+2)} = \frac{n(3n+9)}{2(n+1)(n+2)}$ $a = 3,\ b = 9$	**A1**	cso. Both values, stated or seen in the printed form.
	Alternative: Three linear partial fractions		
	$\frac{6}{r(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+2}$ $\Rightarrow A = \ldots,\ B = \ldots,\ C = \ldots$	**M1**	Writes the term with denominators r, $r+1$, $r+2$ and finds all three constants by any method, cover-up included. Condone a slip in one constant.
	$\frac{6}{r(r+1)(r+2)} = \frac{3}{r} - \frac{6}{r+1} + \frac{3}{r+2}$	**A1**	
	$r = 1:\ \frac{3}{1} - \frac{6}{2} + \frac{3}{3}$ $r = 2:\ \frac{3}{2} - \frac{6}{3} + \frac{3}{4}$ $r = 3:\ \frac{3}{3} - \frac{6}{4} + \frac{3}{5}$ $\vdots$ $r = n-1:\ \frac{3}{n-1} - \frac{6}{n} + \frac{3}{n+1}$ $r = n:\ \frac{3}{n} - \frac{6}{n+1} + \frac{3}{n+2}$	**dM1**	Writes out their partial fractions for at least $r = 1, 2, 3$ and $r = n-1, n$, showing a set of three terms cancelling.
	$\frac{3}{1} - \frac{6}{2} + \frac{3}{2} + \frac{3}{n+1} - \frac{6}{n+1} + \frac{3}{n+2}$ $\left[= \frac{3}{2} - \frac{3}{n+1} + \frac{3}{n+2}\right]$	**A1**	Correct surviving terms, oe, simplified or not.
	$\frac{3(n+1)(n+2)-6(n+2)+6(n+1)}{2(n+1)(n+2)}$ $= \frac{n(\ldots)}{2(n+1)(n+2)}$	**M1**	Combines their $p + \frac{q}{n+1} + \frac{s}{n+2}$ over a common denominator and simplifies the numerator. Condone slips in expanding.
	$\frac{3n^2+9n}{2(n+1)(n+2)} = \frac{n(3n+9)}{2(n+1)(n+2)}$ $a = 3,\ b = 9$	**A1**	cso. Both values, stated or seen in the printed form.
	Total for Question 6	**6**	

Question	Scheme	Marks	Notes
7(a)	$(r^4+4r^3+6r^2+4r+1)-(r^4-4r^3+6r^2-4r+1)$ $=8r^3+8r$	**M1**	Expands both brackets and subtracts, reaching the form pr^3+qr. Condone coefficient slips.
	$\sum_{r=1}^{n}\left((r+1)^4-(r-1)^4\right)=8\sum r^3+8\sum r$ $=(2^4-0^4)+(3^4-1^4)+(4^4-2^4)+\cdots$ $+\left(n^4-(n-2)^4\right)$ $+\left((n+1)^4-(n-1)^4\right)$	**M1**	Sums both sides and writes the left side's terms for at least $r=1,\ 2,\ n-1,\ n$, showing or implying cancellation.
	$=(n+1)^4+n^4-1$	**A1**	Correct surviving terms, oe such as $(n+1)^4+n^4-0^4-1^4$.
	$8\sum r^3=(n+1)^4+n^4-1-8\cdot\frac{1}{2}n(n+1)$ $\Rightarrow\sum r^3=\ldots$	**M1**	Substitutes $\sum r=\frac{1}{2}n(n+1)$ into their equation linking $\sum r^3$, $\sum r$ and n, and makes $\sum r^3$ the subject. Condone sign slips.
	$8\sum r^3=2n^4+4n^3+2n^2=2n^2(n+1)^2$ $\Rightarrow\sum_{r=1}^{n}r^3=\frac{1}{4}n^2(n+1)^2$	**A1 AG**	Correct equation for $8\sum r^3$ simplified to the printed result without error; combined steps accepted.
		[5]	
7(b)	$T_n=\sum_{r=1}^{2n}r^3-\sum_{r=1}^{n}r^3$ $=\frac{1}{4}(2n)^2(2n+1)^2-\frac{1}{4}n^2(n+1)^2$	**M1**	Writes T_n as the sum to $2n$ minus the sum to n, or otherwise in standard sums, and substitutes using (a).
	$=\frac{1}{4}\left(16n^4+16n^3+4n^2-n^4-2n^3-n^2\right)$ $\Rightarrow T_n=\frac{1}{4}n^2(\ldots)$	**dM1**	Expands both terms and collects like powers, reaching the form $\frac{1}{4}n^2(an^2+bn+c)$. Condone arithmetic slips.
	$T_n=\frac{1}{4}n^2(15n^2+14n+3)$	**A1**	Or $a=15$, $b=14$, $c=3$ stated.
		[3]	
7(c)	$\frac{15}{4}$	**B1ft**	ft $\frac{a}{4}$ for their a from (b); oe such as 3.75.
		[1]	
	Total for Question 7	**9**	

Question	Scheme	Marks	Notes
8(a)	$\frac{4r\left(\sqrt{r(r+2)}-\sqrt{r(r-2)}\right)}{r(r+2)-r(r-2)}$	**M1**	Multiplies numerator and denominator by the conjugate, or the RHS by the denominator, and expands as a difference of squares. Condone slips.
	$= \frac{4r\left(\sqrt{r(r+2)}-\sqrt{r(r-2)}\right)}{4r}$ $= \sqrt{r(r+2)} - \sqrt{r(r-2)}$	**A1 AG**	Difference of squares correctly simplified to $4r$, leading without error to the printed result.
		[2]	
8(b)	$r=2:\ \sqrt{8}-\sqrt{0}$ $r=3:\ \sqrt{15}-\sqrt{3}$ $r=4:\ \sqrt{24}-\sqrt{8}$ ... $r=n-1:\ \sqrt{(n-1)(n+1)}-\sqrt{(n-1)(n-3)}$ $r=n:\ \sqrt{n(n+2)}-\sqrt{n(n-2)}$	**M1**	Uses (a) and lists terms for at least $r=2$, $r=3$, $r=n-1$ and $r=n$, with cancellation shown. Condone slips.
	$\sqrt{(n-1)(n+1)}+\sqrt{n(n+2)}-\sqrt{0}-\sqrt{3}$	**A1**	The four surviving terms, in any order; $\sqrt{0}$ may be omitted.
	$= \sqrt{n^2-1}+\sqrt{n^2+2n}-\sqrt{3}$	**A1**	oe, such as $\sqrt{(n-1)(n+1)}+\sqrt{n(n+2)}-\sqrt{3}$, with no zero term remaining.
		[3]	
8(c)	"$\sqrt{6\times 8}+\sqrt{7\times 9}-\sqrt{3}$" $-\sqrt{8}$	**M1**	Substitutes $n=7$ in their (b) and subtracts their $r=2$ term, from (a) or their (b) at $n=2$.
	$= -2\sqrt{2}+3\sqrt{3}+3\sqrt{7}$ $\Rightarrow A=-2,\ B=3,\ C=3$	**A1**	
		[2]	
	Total for Question 8	**7**	

Question	Scheme	Marks	Notes
9(a)	$\frac{1}{(r+a)(r+a+1)(r+a+2)}$ $= k\left(\frac{1}{(r+a)(r+a+1)} - \frac{1}{(r+a+1)(r+a+2)}\right)$	**M1**	Expresses the general term as a multiple of the difference of consecutive terms $\frac{1}{(r+a)(r+a+1)}$, or in partial fractions. Condone a wrong multiple.
	$k = \frac{1}{2}$	**A1**	Correct identity, or correct partial fractions $\frac{1}{2(r+a)} - \frac{1}{r+a+1} + \frac{1}{2(r+a+2)}$.
	$\frac{1}{2}\left[\frac{1}{(a+1)(a+2)} - \frac{1}{(a+2)(a+3)}\right.$ $+ \frac{1}{(a+2)(a+3)} - \frac{1}{(a+3)(a+4)}$ $+ \cdots$ $\left. + \frac{1}{(n+a)(n+a+1)} - \frac{1}{(n+a+1)(n+a+2)}\right]$	**dM1**	Writes their difference for at least $r = 1$, $r = 2$ and $r = n$, showing terms cancelling; with partial fractions, at least $r = 1, 2, 3$ and $r = n-1, n$, showing a set of three cancelling.
	$\frac{1}{2}\left(\frac{1}{(a+1)(a+2)} - \frac{1}{(n+a+1)(n+a+2)}\right)$	**A1**	oe, such as $\frac{n(n+2a+3)}{2(a+1)(a+2)(n+a+1)(n+a+2)}$.
		[4]	
9(b)	$n \to \infty$: $\frac{1}{2(a+1)(a+2)} = \frac{1}{24}$	**M1**	Takes the limit as $n \to \infty$ in their (a), with the terms in n tending to 0, and sets the limit equal to $\frac{1}{24}$.
	$a^2 + 3a - 10 = 0 \Rightarrow a = \ldots$	**dM1**	Rearranges to a three-term quadratic in a and solves it by any means.
	$a = 2$ ($a = -5$ rejected)	**A1**	A0 if $a = -5$ is also given as an answer.
		[3]	
	Total for Question 9	**7**	

Question	Scheme	Marks	Notes
10(a)	$\sum(4r^3 - r) = 4 \cdot \frac{1}{4}N^2(N+1)^2 - \frac{1}{2}N(N+1)$	**M1**	Expands to a polynomial in r and substitutes the standard result for each sum. Condone slips in the expansion.
	$= N^2(N+1)^2 - \frac{1}{2}N(N+1)$	**A1**	Correct expression in N, oe.
	$= \frac{1}{2}N(N+1)\left(2N(N+1) - 1\right)$ $= \frac{1}{2}N(N+1)(2N^2 + 2N - 1)$	**A1 AG**	Correct expression in N leading without error to the printed result; expanding both to the same quartic is accepted.
		[3]	
10(b)	$6r+13 \equiv A(2r+3)+B(2r-1) \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Forms an identity of this form and finds A or B, by substitution or comparing coefficients.
	$\frac{4}{2r-1} - \frac{1}{2r+3}$	**A1**	
	$\frac{4}{(2r-1)2^{r+1}} - \frac{4}{(2r+3)2^{r+3}}$ $4\left[\left(\frac{1}{1\cdot 2^2} - \frac{1}{5\cdot 2^4}\right) + \left(\frac{1}{3\cdot 2^3} - \frac{1}{7\cdot 2^5}\right)\right.$ $+ \left(\frac{1}{5\cdot 2^4} - \frac{1}{9\cdot 2^6}\right) + \cdots$ $+ \left(\frac{1}{(2N-3)2^N} - \frac{1}{(2N+1)2^{N+2}}\right)$ $\left. + \left(\frac{1}{(2N-1)2^{N+1}} - \frac{1}{(2N+3)2^{N+3}}\right)\right]$	**dM1**	Uses their partial fractions with the factor $\left(\frac{1}{2}\right)^{r+1}$, showing terms cancel two apart, from at least $r = 1, 2, 3$ and $r = N-1, N$.
	$\frac{7}{6} - \frac{1}{(2N+1)2^N} - \frac{1}{(2N+3)2^{N+1}}$	**A1**	Correct sum oe, such as $\frac{7}{6} - \frac{6N+7}{(2N+1)(2N+3)2^{N+1}}$. Need not be combined.
		[4]	
10(c)	$N \to \infty$: $\frac{1}{(2N+1)2^N},\ \frac{1}{(2N+3)2^{N+1}} \to 0$	**M1**	Takes the limit as $N \to \infty$ in their sum of the form constant plus terms with a factor 2^{-N}, with those terms tending to 0.
	$\frac{7}{6}$	**A1**	$\frac{7}{6}$, or awrt 1.17.
		[2]	
	Total for Question 10	**9**	

Question	Scheme	Marks	Notes
11(a)	$(m+1)^5-(m-1)^5$, $m=2n-1$ $=(m^5+5m^4+10m^3+10m^2+5m+1)$ $\quad-(m^5-5m^4+10m^3-10m^2+5m-1)$	**M1**	Expands both fifth powers by the binomial theorem, in n or after a substitution such as $m=2n-1$, or expands both sides. Condone sign slips.
	$=10m^4+20m^2+2$ $=10(2n-1)^4+20(2n-1)^2+2$	**A1 AG**	Correct expansions leading without error to the printed identity. If both sides are expanded, both correct and LHS = RHS stated.
		[2]	
11(b)	$\sum_{r=1}^{n}\left((2r)^5-(2r-2)^5\right)$ $=10\sum_{r=1}^{n}(2r-1)^4+20\sum_{r=1}^{n}(2r-1)^2+2n$	**M1**	Replaces n by r in the identity and sums both sides over $r=1$ to n. Condone $\sum 2$ written as 2.
	LHS $=(2^5+4^5+\cdots+(2n)^5)$ $\quad-(0^5+2^5+\cdots+(2n-2)^5)$	**dM1**	Writes out the LHS terms for at least $r=1$, $r=2$ and $r=n$, with the cancellation evident.
	$=(2n)^5=32n^5$	**A1**	$(2n)^5$ oe.
	$\sum_{r=1}^{n}(2r-1)^2=\sum_{r=1}^{n}(4r^2-4r+1)$ $=\frac{4}{6}n(n+1)(2n+1)-\frac{4}{2}n(n+1)+n$	**M1**	Expands $(2r-1)^2$ and substitutes the standard results for $\sum r^2$ and $\sum r$, with $\sum 1=n$; or any valid use of standard results, such as $\sum_{r=1}^{2n}r^2-4\sum_{r=1}^{n}r^2$.
	$=\frac{1}{3}n(2n-1)(2n+1)$	**A1**	Correct expression, oe; need not be factorised.
	$10\sum_{r=1}^{n}(2r-1)^4=32n^5-\frac{20}{3}n(4n^2-1)-2n$ $\Rightarrow\sum_{r=1}^{n}(2r-1)^4=\frac{1}{15}n(48n^4-40n^2+7)$ $\quad=\frac{1}{15}n(4n^2-1)(\ldots)$	**dM1**	Substitutes their $32n^5$ and their cubic for $\sum(2r-1)^2$ into the summed identity, makes $\sum(2r-1)^4$ the subject, and takes out $n(4n^2-1)$ or compares coefficients with the printed form.
	$=\frac{1}{15}n(2n-1)(2n+1)(12n^2-7)$ $a=12$, $b=0$, $c=-7$	**A1**	cso. All three values, stated or seen in the factorised form.
		[7]	
	Total for Question 11	**9**	

Question	Scheme	Marks	Notes
12(a)	$\frac{1}{(3r-2)(3r+4)} = \frac{A}{3r-2} + \frac{B}{3r+4}$ $\Rightarrow A = \ldots,\ B = \ldots\ [= \frac{1}{6}, -\frac{1}{6}]$	**M1**	Uses partial fractions of the form $\frac{A}{3r-2} + \frac{B}{3r+4}$ and finds values for A and B by any method.
	$S_n = \frac{1}{6}\left[(1 - \frac{1}{7}) + (\frac{1}{4} - \frac{1}{10}) + (\frac{1}{7} - \frac{1}{13})\right.$ $+ \cdots + (\frac{1}{3n-5} - \frac{1}{3n+1})$ $\left. + (\frac{1}{3n-2} - \frac{1}{3n+4})\right]$	**dM1**	Writes their terms for at least $r = 1, 2, 3$ and $r = n-1, n$, enough to show the cancellation pattern.
	$S_n = \frac{1}{6}\left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4}\right)$	**A1**	
	$S_n = \frac{5(3n+1)(3n+4) - 4(6n+5)}{24(3n+1)(3n+4)}$	**M1**	Combines their surviving terms, including both in n, over a common denominator. Need not be simplified.
	$= \frac{3n(15n+17)}{24(3n+1)(3n+4)} = \frac{n(15n+17)}{8(3n+1)(3n+4)}$	**A1 AG**	Correct combined fraction from correct surviving terms, reaching the printed result without error.
		[5]	
12(b)	$\frac{5}{24} - S_n = \frac{5}{24} - \frac{1}{6}\left(\frac{5}{4} - \frac{6n+5}{(3n+1)(3n+4)}\right)$	**M1**	Subtracts either form of S_n from (a) from $\frac{5}{24}$ and attempts a single simplified expression.
	$= \frac{6n+5}{6(3n+1)(3n+4)}$	**A1**	oe, such as $\frac{6n+5}{54n^2+90n+24}$.
	$\frac{6n+5}{6(3n+1)(3n+4)} < \frac{1}{k}$ $\Rightarrow k(6n+5) < 54n^2 + 90n + 24$	**M1**	Sets their difference less than $\frac{1}{k}$, clears the fractions and collects to a quadratic inequality in n. Condone slips.
	$54n^2 + (90 - 6k)n + 24 - 5k > 0$	**A1**	oe, any multiple, with like terms collected.
	$n = \frac{6k - 90 \pm \sqrt{(90-6k)^2 - 216(24-5k)}}{108}$	**dM1**	Uses the correct quadratic formula with their coefficients in k, or completes the square, leading to $n = \ldots$.
	$(90 - 6k)^2 - 216(24 - 5k) = 36(k^2 + 81)$ $\Rightarrow n = \frac{k - 15 \pm \sqrt{k^2+81}}{18}$ $k - 15 - \sqrt{k^2+81} < 0$, so as $n \geq 1$ $n > \frac{k - 15 + \sqrt{k^2+81}}{18}$	**A1 AG**	Correct discriminant reaching both roots without error, the region outside the roots, and the lower root rejected with a reason, such as being negative.
		[6]	
	Total for Question 12	**11**	